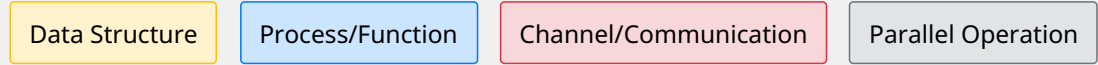


BinaryEntry Workflow - V0.7 Unified Architecture

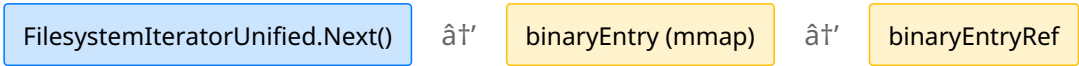
Legend - Data Structures & Terminology



Key Data Structures:

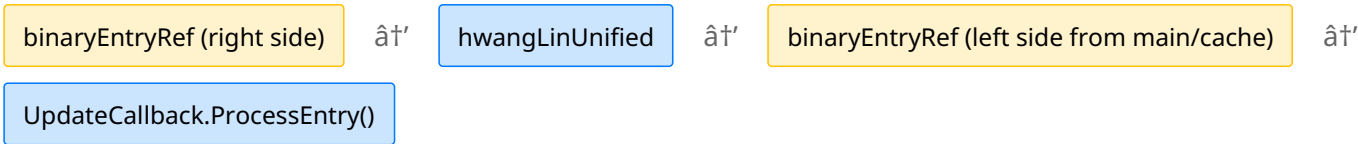
- **binaryEntry**: Raw 68-byte binary structure in mmap memory
- **binaryEntryRef**: Safe pointer wrapper with offset + IndexFile reference
- **BinaryEntryInterface**: Interface for entry access (implemented by binaryEntryRef)
- **skiplistWrapper**: Zero-copy skiplist for ordered entry management
- **parkedSkiplist**: Entries waiting for hash completion, ordered by cookie
- **cookieToEntry map**: Cookie → BinaryEntryInterface lookup for completions
- **hashJobStart**: Hash job submission structure (JobID + file path)
- **hashJobComplete**: Hash completion structure (JobID + computed hash)
- **IoVec**: Vectorio structure for efficient bulk writes

1. Filesystem Scanning



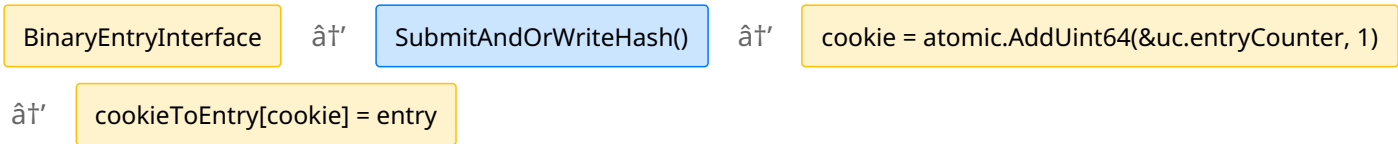
Creates 68-byte entries in scan index mmap, returns safe pointer wrappers

2. Hwang-Lin Comparison



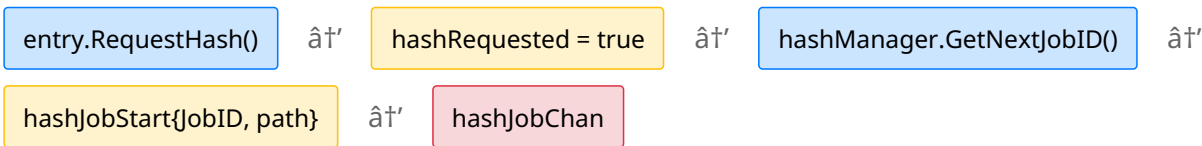
Path-based comparison determines: New, Modified, Unchanged, or Deleted

3. UpdateCallback Processing



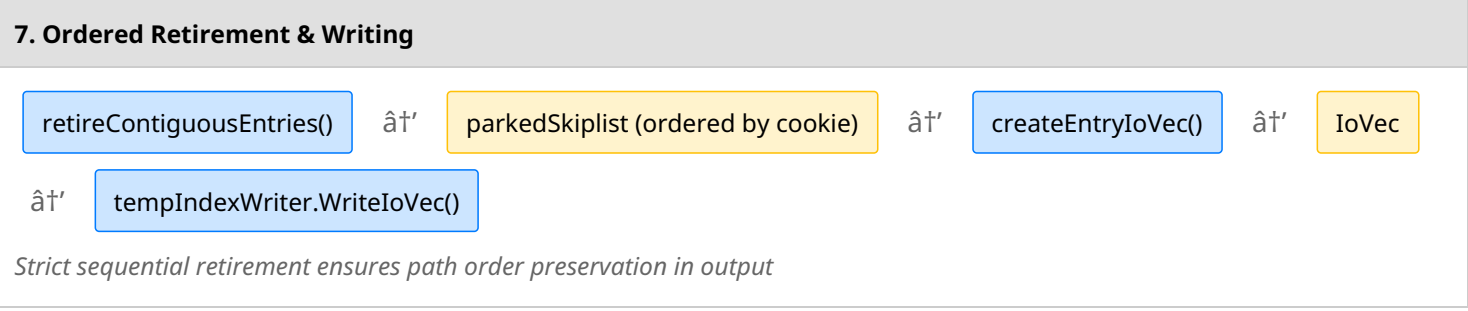
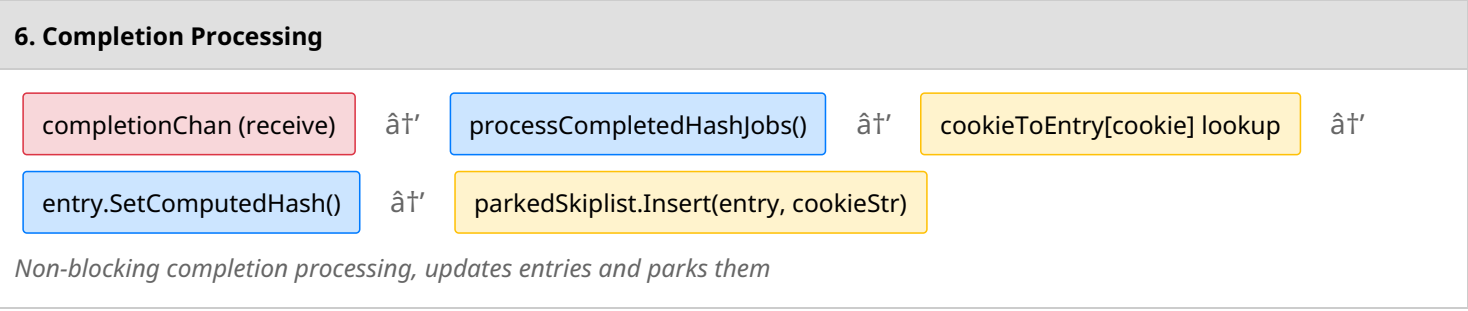
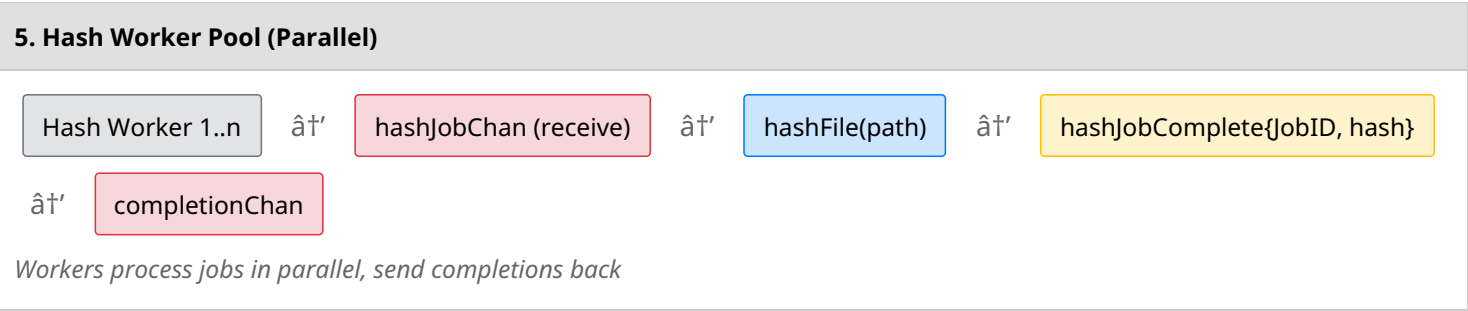
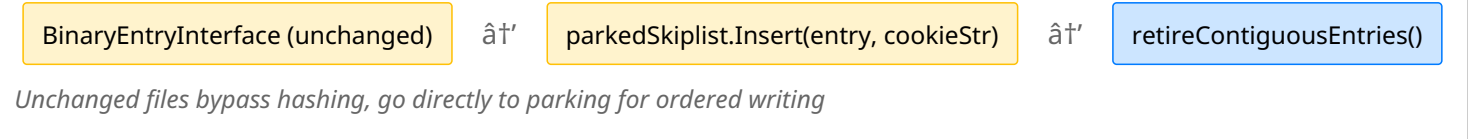
Each entry gets unique cookie for ordering, stored in lookup map

4A. Hash Path (Modified/New Files)



Hash jobs submitted to worker pool with JobID sequence

4B. Direct Write Path (Unchanged Files)



Key Coordination Mechanisms

- **Cookie Sequencing:** Ensures path order preservation through async hash processing
- **Parking Skiplist:** Holds completed entries until sequential retirement is possible
- **Non-blocking Completion:** Prevents goroutine coordination deadlocks
- **JobID Tracking:** Links hash jobs to specific entries via algorithmHashManager
- **Channel Coordination:** hashJobChan â†’ workers â†’ completionChan â†’ parking

Critical Sequencing Rules

- Cookie assignment must be atomic and sequential
- Retirement from parkedSkiplist must be strictly sequential (1, 2, 3, ...)
- Hash completions can arrive out-of-order but writes must be in-order
- Unchanged files bypass hashing but maintain sequence position